

MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

DEPARTMENT OF NETWORKS

Course: BSE4104 Emerging Trends in Software Engineering
Project Title: Makerere Student Support & Onboarding Triage Agent
Academic Year: 2026/2027 | **Semester:** I
Course Convener: Dr. Kamulegeya Grace B. (PhD)
Submission Date: September 4, 2026

Team Composition & Role Allocations

Student Name	Assigned Role	Primary Responsibilities
NAKYANZI FARIDAH	Project & Requirements Lead	Project charter formulation, user personas, scope bounds, and ClickUp task alignment.
KAKOOZA MICHEAL OWEN	Application Lead / DevOps	System architecture design, REST API specifications, backend orchestration, and repository scaffolding.
MPANGA MARVIN JOSEPH	AI Engineering Lead / DevOps	Knowledge corpus assembly, prompt engineering specifications, model integration, and CI/CD pipelines.
LUBEGA LAWRENCE KIRIBWA	Quality & Security Lead	User stories & acceptance criteria, AI Boundary Matrix, test suite design, and safety enforcement.

1. Project Charter (BSE4104 Capstone)

1.1. Minimum Proposal Statement

*"Our system helps **Makerere University students (particularly newly admitted first-years and continuing undergraduates)** complete **academic onboarding, procedural inquiries, and departmental support ticket routing**. AI is used for **semantic retrieval across university regulations, synthesizing procedural step-by-step guidance, and drafting administrative escalation tickets**. Deterministic software remains responsible for **authenticating student identity, checking student enrollment/status records against database tables, and enforcing role-based ticket routing**. The agent may use **approved tools (search_policy_docs, get_student_status, and draft_support_ticket)** but may not **make admissions decisions, alter academic transcripts or grades, waive tuition/functional fees, or grant disciplinary exemptions**. We will build and evaluate the system using **public Makerere University student handbook documents, academic policy manuals, and a curated synthetic dataset of student enrollment profiles and support tickets**."*

1.2. Problem Statement & User Pain Point

Makerere University students encounter recurring operational friction when attempting to navigate academic regulations, semester registration timelines, accommodation guidelines, course transfer requirements, and missing examination mark reporting. Pertinent information is fragmented across extensive policy manuals, departmental notices, and portal guidelines. Consequently, students face extensive queues at College and School Registrars' offices or rely on inaccurate peer advice. Furthermore, academic desk officers are overloaded with repetitive first-line informational queries, delaying the resolution of critical student escalations.

1.3. Target Personas

- **Undergraduate / First-Year Student:** Requires prompt, grounded procedural guidance and automated drafting of formal administrative inquiries tailored to their academic program.
- **College Registrar / Helpdesk Desk Officer:** Requires structured, pre-triaged support tickets containing verified student profile context and explicit policy clause references.

1.4. Value Proposition & AI Justification

- **AI Value:** Enables semantic understanding of varied natural language phrasing, contextual knowledge retrieval from unstructured PDF policies, multi-step intent breakdown, and concise ticket drafting.
- **Deterministic Bounds:** Retains strict deterministic control over authentication, database lookups for student holds, schema validation, and rate limiting, guaranteeing system integrity.

1.5. Scope & Safety Constraints

- **In-Scope:** Grounded policy Q&A from an indexed Makerere handbook corpus (10–50 documents), mock profile queries, automated guidance generation, and human-confirmed support ticket drafting.
- **Prohibited Actions:** Direct database writes to marks/fees, autonomous admission approvals, granting disciplinary exemptions, or open-ended external web browsing.
- **Data Feasibility:** Built entirely on public Makerere University academic regulations, joining instructions, college handbooks, and synthetic student database records.

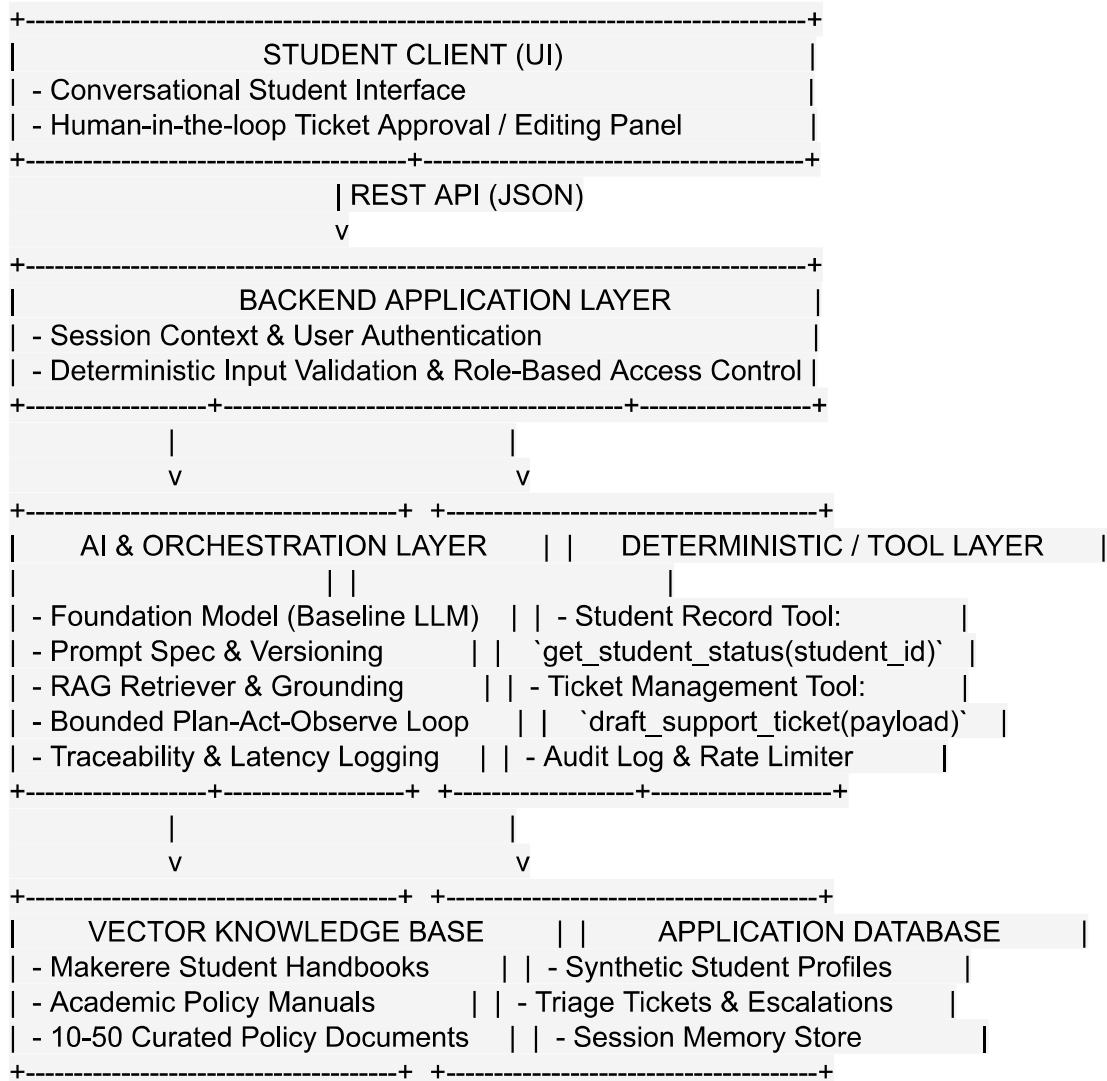
2. AI Boundary Matrix

Operational Area	AI Autonomous Layer	Deterministic Software Layer	Human Approval / In-The-Loop
Policy Retrieval	Semantic matching across regulations; source citation extraction.	Document indexing boundaries; access control to document stores.	Student inspects provided source citations.
Procedural Guidance	Synthesizes step-by-step instructions (e.g., examination appeals).	Schema validation of input queries and parameters.	Student validates generated guidance steps.
Student Status Verification	Generates tool parameters (e.g., student_id).	Read-only database lookup; session authentication validation.	Student confirms accuracy of displayed profile data.
Ticket Drafting & Routing	Classifies department; drafts structured inquiry summary.	Database write to ticket queue; rate limiting & schema enforcement.	Mandatory: Student reviews, edits, and submits ticket.
Academic Actions (Marks/Fees)	<i>Strictly Prohibited</i>	Immutable ledgers; read-only constraints on academic databases.	Mandatory: Academic Registrar / School Dean full authority.

3. User Stories & Acceptance Criteria

1. **US-01 (Orientation & Registration):** As a first-year student, I want to query physical registration requirements in natural language, so that I complete my enrollment before official deadlines.
 - *Acceptance Criteria:* The agent cites the Makerere Joining Instructions and returns an actionable 3-step checklist.
2. **US-02 (Missing Examination Mark Escalation):** As a continuing student, I want to know the official procedure for reporting a missing grade, so that I file an appeal with the correct department.
 - *Acceptance Criteria:* The agent extracts Examination Regulation clauses, specifies the 30-day reporting window, and lists required forms.
3. **US-03 (Profile Status Check):** As a student, I want the agent to inspect my active enrollment and holds, so that I receive advice tailored to my specific program.
 - *Acceptance Criteria:* Given a valid student_id, the agent calls get_student_status and contextualizes responses with college/program data.
4. **US-04 (Retake Policy Guidance):** As a student, I want to understand retake fee computations and credit unit limits, so that I plan my semester course load.
 - *Acceptance Criteria:* The agent quotes official retake policies and rejects any request to alter transcript records.
5. **US-05 (Draft Escalation Ticket):** As a student facing a timetable clash, I want the agent to draft a formal ticket to the School Timetable Coordinator, so that I do not write it manually from scratch.
 - *Acceptance Criteria:* The agent drafts an editable ticket containing student ID, conflicting course codes, and policy references.
6. **US-06 (Human-in-the-Loop Confirmation):** As a student, I want to review drafted tickets prior to filing, so that inaccurate data is never submitted on my behalf.
 - *Acceptance Criteria:* Tickets are persisted to the database only after the student triggers an explicit "Confirm & Submit" action in the UI.
7. **US-07 (Unsupported Query Handling):** As a system auditor, I want the agent to state when a query is outside university policies, so that hallucinations are prevented.
 - *Acceptance Criteria:* Out-of-scope questions return a standard fallback response acknowledging lack of grounded data.
8. **US-08 (Change of Program Rules):** As a student, I want to view transfer cut-off criteria, so that I determine eligibility for course switching.
 - *Acceptance Criteria:* The agent retrieves cut-off rules, application windows, and Academic Registrar routing instructions.
9. **US-09 (Helpdesk Queue Sorting):** As a College Registrar, I want to view pre-triaged tickets with policy tags, so that incoming cases are resolved efficiently.
 - *Acceptance Criteria:* Helpdesk views display tickets sorted by college/department with verified policy citations.
10. **US-10 (Adversarial Safety Enforcement):** As a systems administrator, I want the agent to reject prompts attempting to alter marks or waive fees, so that university integrity is enforced.
 - *Acceptance Criteria:* Prompts demanding automated grade modifications or fee waivers trigger explicit security refusals.

4. Initial Architecture & Component Interaction



5. Project Infrastructure & Task Allocation

5.1. Standard Repository Directory Layout

```
makerere-student-support-agent/
├── README.md
├── .env.example
├── docs/
│   ├── requirements/      # Project charter, user stories, boundary matrix
│   └── architecture/     # Context and sequence diagrams
```

weekly-reports/	# Week 1 - Week 8 progress reports
evaluation/	# Scenario logs and evaluation tables
prompts/	# Versioned prompt specifications (v1.0+)
knowledge/	# Document register & provenance metadata
src/	# Application & agentic orchestration code
tests/	# Unit, integration, and prompt evaluation tests
evidence/	
traces/	# Agent execution & tool traces
screenshots/	# ClickUp & system execution captures
demo/	# Presentation assets & slide decks

5.2. ClickUp Week 1 Task Allocations

- **Task 101:** Project Charter & Minimum Proposal Statement (Owner: *NAKYANZI FARIDAH*)
- **Task 102:** User Stories & AI Boundary Matrix (Owner: *LUBEGA LAWRENCE KIRIBWA*)
- **Task 103:** System Architecture & Context Diagram (Owner: *KAKOOZA MICHEAL OWEN*)
- **Task 104:** Corpus Collection & Source Register (Owner: *MPANGA MARVIN JOSEPH*)
- **Task 105:** GitHub Repository Scaffolding & CI Pipeline (Owners: *MPANGA MARVIN JOSEPH & KAKOOZA MICHEAL OWEN*)

6. Week 1 Progress Report

Group ID: BSE4104 Group Capstone Team

Week Ending: September 4, 2026

6.1. Work Completed

- Finalized project domain: University Student-Support Case Agent tailored to Makerere University.
- Authored Project Charter, AI Boundary Matrix, and 10 testable user stories with explicit acceptance criteria.
- Designed system architecture diagram cleanly decoupling AI orchestration from deterministic validation.
- Initialized Git repository scaffold and configured ClickUp milestone tracking.

6.2. Key Engineering Decisions

- **Deterministic Human Gate:** Mandated explicit human-in-the-loop confirmation before ticket submission.
- **Corpus Boundary:** Restricted knowledge store to public Makerere policies and synthetic student records to ensure zero proprietary data dependencies.

6.3. Challenges & Mitigations

- **Challenge:** Risk of system being evaluated as a simple conversational wrapper.

- **Mitigation:** Bounded execution to a deterministic multi-step workflow requiring tool lookups and draft confirmations.

6.4. Individual Accountability Summary

- **NAKYANZI FARIDAH:** Project Charter formulation, persona definitions, and scope boundaries.
- **MPANGA MARVIN JOSEPH:** Knowledge corpus specification, baseline prompt architecture, and repository CI scaffolding.
- **KAKOOZA MICHEAL OWEN:** System architecture, REST API design, and repository layout.
- **LUBEGA LAWRENCE KIRIBWA:** User stories authoring, acceptance criteria verification, and safety boundary definition.

6.5. Plan for Week 2

- Integrate baseline foundation model and produce Model Selection Note.
- Author Prompt Specification v1.0 and execute the 10-case prompt evaluation suite.